

AppliedScientist: Automated Scientific Revision Through Iterative AI Reviewing

Anonymous submission

Abstract

Automated reviewing systems are increasingly evaluated based on the quality of the reviews they produce. Yet a review is only useful if acting on it leads to a measurable improvement in the paper. We present *AppliedScientist*, a closed-loop system that couples an autonomous AI scientist with an AI reviewer, and evaluate it by iteratively revising rejected papers from a range of research subfields. To mirror how human authors build on earlier drafts, the AI scientist has access to its previous versions during revision. To avoid bias from prior judgments, however, each review is generated independently, with the reviewer having no memory of earlier feedback or scores. We compare three revision settings: one initialized with the original venue reviews, one initialized with AI-generated reviews, and autonomous self-revision using the same fixed prompt in every round. Because the reviewer both guides and evaluates the revision, we also assess the human-initialized revisions using Stanford Reviewer as an independent evaluator. Reviewer-guided revision consistently improves more than fixed-prompt self-revision, and Stanford Reviewer also assigns higher scores to later revisions. *AppliedScientist* resolves 128 of 150 execution-related weaknesses (85.3%), but only 2 of 18 idea-related weaknesses (11.1%), suggesting that iterative revision is effective at improving experiments and implementation, but rarely changes concerns about novelty or significance.

Introduction

Scientific publishing is an iterative process rather than a one-time evaluation. Reviewers identify strengths and weaknesses, and authors revise the work before it is considered again. Addressing reviewer feedback, however, often requires substantially more than revising the manuscript. Authors may need to reproduce results, run new experiments, evaluate stronger baselines, and determine whether the resulting evidence addresses the reviewers’ concerns. As language models become increasingly capable of conducting research and reviewing scientific papers, a natural question emerges: Can this process of scientific review and revision itself be automated?

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Recent work has steadily automated different stages of the scientific publication process. AI scientists can conduct research and prepare manuscripts (Lu et al. 2024; Yamada et al. 2025; Gottweis et al. 2026), while AI reviewers can produce detailed critiques grounded in a submitted paper (Zhu et al. 2025; Goyal et al. 2026; Jiang and Ng 2026). Yet generating a paper or a review does not establish that acting on the review improves the underlying research. In a closed revision loop, the reviewer is not merely an evaluator; it supplies the feedback signal that determines which weaknesses the scientist addresses, much as a grader or reward model directs optimization in reinforcement learning. Even a capable scientist can therefore optimize toward the wrong objective when its reviewer is unreliable. Existing systems provide either a broad experimental action space under fixed human feedback or dynamic AI feedback restricted to paper generation or textual rewriting. What remains unclear is whether an independently evaluated AI reviewer can repeatedly guide code, experiment, and manuscript revisions of the same existing paper without replacing its central contribution.

We study this question using real rejected papers, which already contain an implemented method, experimental code, and reviewer feedback describing the work’s shortcomings. We introduce **AppliedScientist**, a closed-loop system in which a scientist revises the implementation, executes experiments, analyzes the resulting artifacts, and updates the manuscript. After each revision, an independent **AI Reviewer** evaluates only the current manuscript and returns fresh feedback for the next round. The scientist retains its previous code, results, manuscripts, and feedback so that revisions accumulate and the reviewer retains no history, preventing earlier judgments or scores from biasing its assessment of the current version. The central research contribution remains fixed, allowing us to study revision rather than replacement of the original project.

We evaluate three sources of revision guidance. In the *human-initialized* condition, written venue reviews guide the first round and the AI Reviewer guides subsequent rounds. In the *AI-initialized* condition, the AI Reviewer also supplies the first-round feedback. In *autonomous self-revision*, the scientist instead receives the same fixed self-review prompt in every round. The scientist has the same research capabilities in all three conditions, isolating the effect of the guidance it receives. Because our AI Reviewer both guides and scores

the reviewer-driven conditions, we additionally use Stanford Reviewer to evaluate every saved version from the human-initialized trajectories.

We evaluate AppliedScientist on 25 rejected and 5 borderline-accepted ICLR papers spanning multiple research domains. Reviewer-guided revision improves substantially more than autonomous self-revision, while Stanford Reviewer also assigns higher scores to later human-initialized versions. The effect depends strongly on the type of weakness: AppliedScientist resolves 128 of 150 execution-related weaknesses (85.3%), but only 2 of 18 idea-related weaknesses ($\sim 11.1\%$). Iterative revision can therefore strengthen the implementation, experimental evidence, and presentation of an existing contribution, but it rarely resolves concerns about the novelty or significance of the contribution itself

Our contributions are three-fold:

- We introduce AppliedScientist, an iterative scientist-reviewer system that revises an existing paper’s implementation, experiments, and manuscript in response to fresh reviews of each version.
- We treat the reviewer as a first-class component: we benchmark its review quality separately, compare its dynamic guidance with fixed-prompt self-revision, and use an external reviewer to evaluate the resulting trajectories.
- We characterize what autonomous revision can and cannot improve: execution-related concerns can often be resolved through systematic revision, whereas novelty-related concerns remain substantially more difficult.

Related Work

AI Scientists and Autonomous Research Autonomous AI systems increasingly cover multiple stages of the scientific research lifecycle, including idea generation, literature review, experimentation, manuscript preparation, and peer review (Lu et al. 2024; Yamada et al. 2025; Gottweis et al. 2026; Baek et al. 2025; Weng et al. 2025; Ghareeb et al. 2026; Mitchener et al. 2025; Schmidgall et al. 2025). Prior systems already recognize the value of reviewing, but use it in different ways. AI Scientist evaluates a completed manuscript and archives the result for future research generations, AI Scientist-v2 uses VLM critiques to refine experimental and manuscript figures, and CycleResearcher uses CycleReviewer scores to construct preference pairs across training rounds (Lu et al. 2024; Yamada et al. 2025; Weng et al. 2025). Robin and Kosmos push the same loop into new domains: Robin runs a full hypothesis-generate-and-test cycle for experimental biology and Kosmos shares a structured model across parallel analysis and literature-search agents to sustain longer, coherent discovery sessions (Ghareeb et al. 2026; Mitchener et al. 2025). These works establish strong foundations for autonomous scientific research. In contrast, our work focuses on iterative scientific revision guided by repeated peer review.

Automated Peer Review Automated reviewers increasingly use structured reasoning, multiple agents, and literature retrieval to produce grounded critiques (Zhu et al. 2025; Goyal et al. 2026; Jiang and Ng 2026; Idahl and Ahmadi

2024; Taechoyotin and Acuna 2025; Weng et al. 2026). These systems differ mainly in how they get to a grounded critique: OpenReviewer fine-tunes directly on expert reviews, Remor trains with reinforcement learning against a reward function built from human preferences, and DeepReviewer 2.0 forces every claim in its output to be traceable to a specific location in the manuscript before it is released (Idahl and Ahmadi 2024; Taechoyotin and Acuna 2025; Weng et al. 2026). Goyal et al. (2026), Jin et al. (2024), D’Arcy et al. (2024), Jiang and Ng (2026) and Tan et al. (2024) focus on the use of multiple agents, distinct personas, and retrieval mechanisms to simulate collaborative reviewing. This literature primarily evaluates the review itself. We separately benchmark our reviewer, and then observe whether acting on its feedback improves the research.

AI-Assisted Scientific Revision Compared with automated reviewing, autonomous scientific revision remains relatively underexplored. The closest systems each cover one side of our setting. Apres revises existing scientific manuscripts by optimizing a learned rubric predictive of future citation impact, but restricts its action space to presentation and protects experimental results from modification (Zhao et al. 2026). CycleResearcher studies the author-reviewer interaction during automated paper generation, but operates on synthetically generated manuscripts and does not execute new experiments during refinement (Weng et al. 2025). Alongside these systems, recent resources have emerged for studying revision itself. Luo et al. (2026) benchmark an LLM’s ability to revise papers using real reviewer feedback, allowing repository edits, experiment execution, and manuscript revision, but acts once on a fixed set of human weaknesses rather than obtaining a new review of the revised paper. Other systems stay at the level of text entirely. ARIES aligns reviewer comments to the edits they prompted in the original manuscript, without generating new edits itself, and XtraGPT, trained on 140K instruction–revision pairs derived from published papers, rewrites one section at a time (e.g. the introduction) to satisfy a stated instruction, without touching the underlying experiments (D’Arcy et al. 2024; Chen et al. 2026). Together, these works establish revision as an emerging research direction, although they primarily focus on textual revision rather than experimentally validating reviewer feedback.

Methodology

AppliedScientist consists of two components: a scientist that revises the implementation, experiments, and manuscript, and an AI reviewer that independently reviews each revision and provides feedback for the next iteration. We describe each component below.

AppliedScientist Revising an existing scientific paper differs fundamentally from generating one from a new research idea. During research generation, the problem formulation, methodology, and experimental design may all evolve as the work develops. Revision instead begins with an established contribution, implementation, and manuscript, where the goal is to address reviewer concerns without replacing

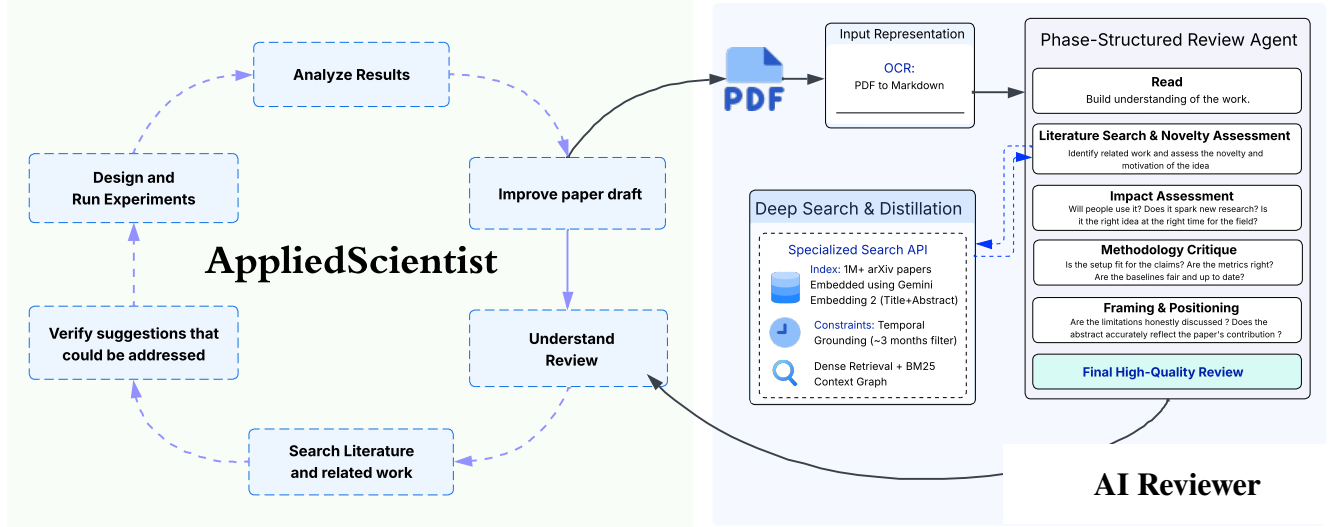


Figure 1: Overview of AppliedScientist. The scientist iteratively verifies reviewer suggestions, runs experiments, analyzes results, and updates the manuscript. The review component independently reads each revision, searches the literature, assesses its technical quality and significance, and returns structured feedback for the next round.

the central research idea. AppliedScientist is designed for this constrained setting.

Given a rejected paper, its source repository, and current feedback, the scientist is instructed to address every reviewer concern while determining how each should be resolved. Depending on the feedback, this may require inspecting the repository and manuscript, searching the literature, editing implementation, reproducing results, adding baselines and presenting ablations, executing new experiments and analyzing the results and updating the manuscript. The central research contribution is treated as fixed; if addressing a concern would require changing that contribution, the scientist records it as unresolved rather than reframing the work as a different project.

Throughout the revision process, the scientist maintains the complete revision history, including previous manuscript versions, reviewer feedback, experimental results, and intermediate analyses. Each revision builds upon the results and decisions from earlier iterations rather than restarting from the original submission. The revision procedure is otherwise identical across all three conditions; only the guidance provided to the scientist changes.

Review Component Our reviewer is inspired by how human reviewers evaluate scientific papers. Rather than assessing every aspect of a paper simultaneously, reviewers gradually build an understanding of the work before forming specific judgments. For example, novelty cannot be assessed without understanding the relevant literature, and experimental results cannot be evaluated without first understanding the claims they are intended to support. Skipping or reordering these steps can lead to unreliable assessments. Our reviewer therefore follows a structured review process in which each stage builds upon the information established by the previous one.

The reviewer first identifies the paper’s contributions, methodology, and evaluation setting before conducting an independent literature search beyond the cited references. To assess novelty, it performs a deep search over an indexed collection of around one million arXiv papers. The search retrieves literature from multiple perspectives, including direct competitors, alternative approaches to the same problem, recent benchmark methods, and work built upon similar baselines. It then evaluates the paper’s novelty, positioning with respect to prior work, the adequacy of its experimental evaluation, and its broader impact before synthesizing these observations into a structured review containing strengths, weaknesses, and suggestions for the next revision.

Unlike the scientist, our reviewer intentionally maintains no memory of previous revisions. Every manuscript version is reviewed independently without access to earlier drafts, reviewer comments, or historical scores. Retaining this history could bias the review by allowing judgments to be influenced by previous evaluations or previous scores rather than the quality of the submitted manuscript. Before producing the final review, the reviewer performs a verification pass to ensure that factual statements and comparisons are supported by either the manuscript or the retrieved literature.

Revision Loop Let V_0 and C_0 denote the original manuscript and codebase. At revision round t , the scientist receives the previous manuscript and codebase (V_{t-1}, C_{t-1}), its accumulated revision history, and guidance G_{t-1} . It then updates the implementation, performs any required experiments, and produces revised code, experimental artifacts, and manuscript (C_t, E_t, V_t). The reviewer evaluates only V_t (together with any literature retrieved during that review) and generates a new review R_t without access to earlier manuscript versions, reviews, or evaluation scores. In the reviewer-guided conditions, $G_t = R_t$ for the next revision

Domain	N	Decision	
		Rejected	Borderline
NLP	12	11	1
Computer Vision	4	3	1
Graph Learning	5	4	1
Time Series	5	4	1
Reinforcement Learning	4	3	1
Total	30	25	5

Table 1: We curate 30 ICLR papers spanning five research domains and retain only papers with publicly available source code that can be reproduced on a single GPU within our VRAM budget.

round. In the autonomous self-revision condition, G_t remains the same fixed self-review prompt, while R_t is used only for evaluation.

Experimental Setup

Data We mine 25 rejected and 5 borderline-accepted ICLR papers with both publicly available reviews and code. To ensure a diverse evaluation set, we sample papers spanning multiple research domains while preferring those with low variance in original human reviewer scores (Table ??). Finally, we filter papers by computational requirements, retaining only those that can be executed on CPU or a single GPU with 96 GB of VRAM.

Setup We evaluate AppliedScientist under the three revision conditions summarized in Table 2. The human-initialized condition begins with the original venue reviews, whereas the AI-initialized condition replaces them with reviews generated by our reviewer. In both cases, our reviewer evaluates every subsequent revision, and each new review guides the next iteration. By contrast, the autonomous self-revision condition uses no external reviewer feedback and the scientist relies on a same self-review prompt throughout. This comparison isolates the effect of iterative reviewer feedback from the scientist’s ability to revise autonomously. For all conditions, V_0 denotes the original manuscript and V_1 – V_5 denote successive revisions. Each revision run takes approximately 9 hours on a 96 GB VRAM GPU.

Evaluation Our reviewer scores every saved manuscript version without access to earlier drafts, feedback, or scores. Because this reviewer also guides the reviewer-driven conditions, we separately score the human-initialized trajectory with Stanford Reviewer. Stanford Reviewer is used only for external evaluation and its outputs are never shown to the scientist. The score and threshold analyses below use the 30 completed human-initialized trajectories.

Reviewer Evaluation

In a closed revision loop, the reviewer serves both as an evaluator and as the feedback signal that guides the scientist’s next action. Reviewer validity is therefore a prerequisite for interpreting downstream improvements: a stronger scientist

does not make the loop reliable if its reviewer consistently rewards the wrong changes. Before using our reviewer’s scores to assess revision, we compare it with existing automated reviewers using the evaluation setup introduced in DeepReviewBench and ScholarPeer (Zhu et al. 2025; Goyal et al. 2026). We evaluate on 300 papers from 2020–2026, using the ICLR 2024 and ICLR 2025 splits of DeepReviewBench together with 300 papers from the AgentReviewer test set containing ICLR papers from 2020–2023. To evaluate performance on newer research, we additionally include a held-out set of 50 ICLR 2026 submissions. We adopt the extensions to the original DeepReviewBench protocol, replace the plausibility criterion with significance assessment, and allow the judge to use web search for novelty verification, following ScholarPeer.

We use a search-enabled judge that receives the input paper together with reviews from two anonymized systems, with review order randomized to avoid positional bias. The judge compares the reviews across five dimensions: Technical Accuracy, Constructive Value, Analytical Depth and Significance Assessment, with strong penalties for hallucinated claims and generic feedback. Table 3 shows that our Reviewer outperforms all baselines across nearly every evaluation dimension. On Constructive Value, however, it achieves performance comparable to Stanford Agent Reviewer (44.3% vs. 45.7% win rate), with the remaining 10% of comparisons resulting in ties.

Across these papers, our Reviewer correlates with mean human ratings at $\rho = 0.571$ with DeepSeek V4 Flash and matches the venue’s decision 76.2% of the time, against $\rho = 0.48$ and 67.2% with MiniMax M2.7 (Figure 3). The Stanford Reviewer falls between the two, at $\rho = 0.51$ and 71.1%.

Results and Discussion

Does Reviewer Guidance Improve Papers? We first evaluate the human-initialized condition, in which the original written venue reviews guide V_1 and fresh feedback from our reviewer guides each later revision. Reviewer scores improve steadily across successive rounds. We compare this condition with autonomous self-revision, in which the scientist receives the same fixed self-review prompt in every round and no reviewer-generated feedback (Figure 4). Although the scientist retains the same literature search, code execution, experiment design, analysis, and manuscript-editing capabilities, removing iterative reviewer feedback substantially reduces improvement. This suggests that the gains arise from the iterative scientist-reviewer process rather than from the scientist alone.

We also evaluate the AI-initialized condition, in which our reviewer generates the initial feedback on V_0 instead of using the venue reviews. This condition consistently outperforms autonomous self-revision and converges to nearly the same final score as the human-initialized condition after five revision rounds.

Figure 2 shows trajectories for the human-initialized condition across different scientific domains. The upward pattern appears in every represented domain rather than being confined to one research area.

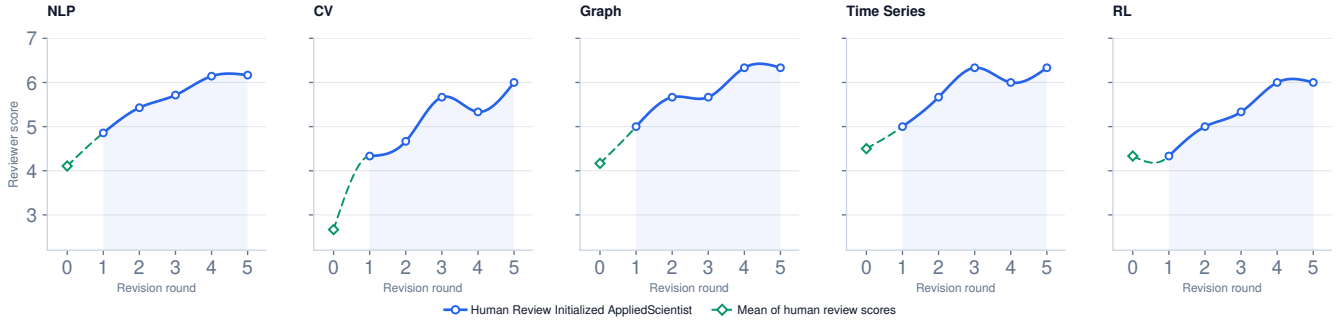


Figure 2: Score trajectories across five scientific domains for the human-initialized condition. Round 0 reports the mean score assigned to the original manuscript by the human venue reviewers. AppliedScientist uses the written venue reviews to produce Round 1. Thereafter, our AI reviewer independently reviews each revision, and its feedback guides the following round.

Condition	Guidance used for V_1	Guidance used for V_2-V_5	Role of our reviewer
Human-initialized	Original written venue reviews of V_0	A fresh, independent AI review of the preceding version	Guides V_2-V_5 and scores each version
AI-initialized	A fresh, independent AI review of V_0	A fresh, independent AI review of the preceding version	Guides V_1-V_5 and scores each version
Autonomous self-revision	Fixed self-review prompt	The same fixed self-review prompt in every round	Evaluation only; no feedback is shown to the scientist

Table 2: Revision conditions. Guidance is the feedback shown to the scientist before it produces the indicated version. Evaluation-only reviews and scores are never exposed to the scientist. Stanford Reviewer is applied only to the saved versions from the human-initialized condition.

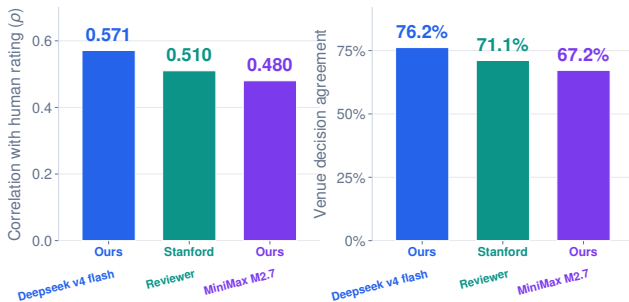


Figure 3: Score alignment with venue reviews on a set of 650 ICLR 2020-26 papers. Left: Spearman correlation with the mean human rating. Right: agreement with the venue’s accept/reject decision.

External Evaluation. To test whether the improvements in our main human-initialized condition are specific to our reviewer, we separately evaluate every saved version from that condition using Stanford Reviewer. We do not run Stanford Reviewer on the AI-initialized or autonomous self-revision conditions, and its evaluation-only reviews are never provided to the scientist. Although Stanford Reviewer assigns different absolute scores, it observes the same upward trajectory across the human-initialized revisions, indicating that the improvements produced by the AppliedScientist with AI Reviewer reflect genuine gains in paper quality rather than optimization towards a single reviewer.

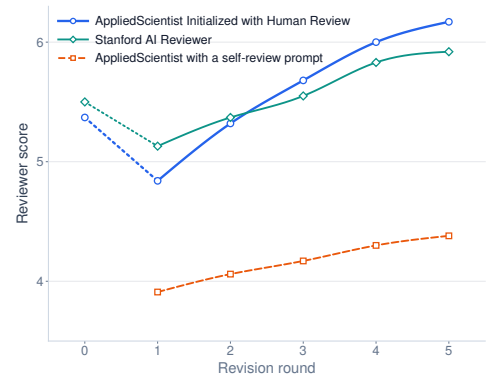


Figure 4: Under our reviewer, the human-initialized condition improves substantially more than autonomous self-revision with a fixed prompt. Stanford Reviewer, applied only to the human-initialized trajectory, also assigns higher scores to later revisions.

Score Improvements In the human-initialized condition, our reviewer scores the original papers at 5.37 on average and the best revision for each paper at 6.53, a gain of +1.16 points. Under the same best-revision aggregation, Stanford Reviewer increases from 5.50 to 6.15, a gain of +0.65 points. At their best revision, all 30 papers score at least 6 under our reviewer, compared with only 12 at V_0 (Table 4).

Category	Baseline	Technical Accuracy		Constructive Value		Analytical Depth		Significance Assessment	
		Win (%)	Loss (%)	Win (%)	Loss (%)	Win (%)	Loss (%)	Win (%)	Loss (%)
Fine-tuned	CycleReviewer-8B	100	0	100	0	99	1	100	0
	CycleReviewer-70B	99.2	0.4	100	0	100	0	99.6	0.2
	DeepReviewer-7B	96.5	2.1	99.4	0.4	99.2	0.6	97.8	1.1
	DeepReviewer-14B	91.2	3.7	96.0	2.2	94.7	2.6	97.9	1.3
LLM	Gemini 3.1 Flash	60.6	21.5	88.2	6.6	75.1	9.9	74.2	4.6
	Gemini 3.1 Pro	62.2	11.7	83.9	6.5	71.6	9.0	70.7	5.1
	DeepSeek V4 Flash	74.3	10.9	92.5	7.1	78.9	7.4	82.3	3.9
	DeepSeek V4 Pro	78.6	7.5	96.5	5.0	79.2	5.1	80.4	3.2
LLM w/ Search	Gemini 3.1 Flash	55.8	12.8	85.7	8.5	70.5	10.2	72.8	6.3
	Gemini 3.1 Pro	56.3	14.6	80.1	7.6	68.4	8.8	66.9	5.8
	DeepSeek V4 Flash	71.4	15.3	89.4	7.0	76.9	7.5	80.1	5.0
	DeepSeek V4 Pro	72.9	17.6	93.1	8.5	73.8	8.1	79.7	6.4
Multi-Agent Framework w/ Search	Agent Review (Gemini 3.1 Pro)	52.1	18.9	89.5	10.4	60.7	10.8	65.6	7.9
	Agent Review (DeepSeek V4 Flash)	61.7	20.5	95.3	11.2	72.8	11.6	68.5	8.7
	AI Scientist v2 (Gemini 3.1 Pro)	50.2	22.3	84.5	12.1	58.5	12.5	62.2	9.5
	AI Scientist v2 (DeepSeek V4 Flash)	56.7	24.1	94.2	13.2	88.3	13.6	67.1	10.3
	DeepReviewer-v2 (StepFun 3.5 Flash)	63.4	16.8	95.6	15.0	91.8	15.4	77.3	11.6
	Stanford Agent Reviewer*	48.5	24.8	44.3	45.7	52.7	14.0	54.6	10.8

Table 3: Pairwise preference evaluation of our AI Reviewer against existing automated reviewers on 650 ICLR papers from 2020-26. A search-enabled LLM judge compares anonymized review pairs using the evaluation rubric of Goyal et al. (2026). Entries report the percentage of comparisons won by our reviewer (Win) or the baseline (Loss), with the remaining comparisons resulting in ties. Higher Win and lower Loss indicate better performance. * Stanford Reviewer was evaluated on 100 papers because it is not open-source and is accessible only through a web interface.

Score $\geq$	Papers at V_0	Papers at $V_{\max}$	Δ
Our reviewer			
5	24/30	30/30	+6
6	12/30	30/30	+18
7	3/30	14/30	+11
Stanford Reviewer			
5.0	28/30	30/30	+2
5.75	8/30	25/30	+17
6.0	3/30	19/30	+16

Table 4: Number of papers scoring at or above each threshold, before revision (V_0) and at each paper’s best revision ($V_{\max}$), out of 30 papers.

What Can Revision Improve? To better understand the limits of autonomous revision, we ask a simple question: What kinds of reviewer criticisms can revision actually resolve? To answer this, we first analyse why papers are rejected. We randomly sample 500 rejected ICLR papers and use Gemini 3.1 Pro to classify every reviewer criticism as either an *execution* issue (e.g., missing baselines, weak evidence, unsupported claims, or unclear writing) or an *idea* issue (e.g., insufficient novelty or significance). This analysis is used only to estimate the distribution of rejection reasons and is separate from our evaluation set. Execution issues account for 48% of rejections, idea-related issues for 35%, and

the remaining 17% involve both.

Having identified these two broad categories, we next ask which of them can actually be resolved through iterative revision. On our evaluation set, AppliedScientist resolves 128 of 150 execution weaknesses (85.3%), but only 2 of 18 idea weaknesses ($\sim 11.1\%$). As shown in Figure 5, revision consistently succeeds on suggestions that require stronger evidence via additional experiments, baselines, ablations, or clearer presentation, but rarely changes reviewer judgements about novelty or significance.

This difference persists across revision rounds. Our reviewer retains novelty-related objections until the final round, often citing prior work to justify them. Once the contribution itself is judged to lack novelty, additional evidence seldom changes the outcome. Our results therefore suggest that autonomous revision is well suited to strengthening the evidence behind an idea, but improving the idea itself requires returning to the ideation stage rather than continuing the revision loop.

Conclusion

We presented AppliedScientist, a closed-loop system for iteratively revising rejected research papers using repeated AI reviewer feedback. Across 30 ICLR papers, reviewer-guided revision consistently improved papers more than autonomous self-revision with a fixed prompt, with most improvements coming from stronger experiments, implementation, evaluation, and presentation rather than changes to the underlying

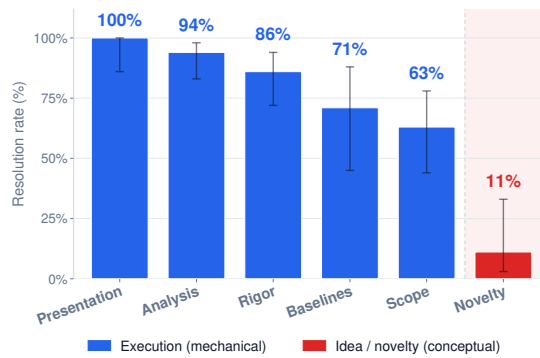


Figure 5: Resolution rate of AI-suggested weaknesses by AppliedScientist. Most empirical weaknesses are resolved, whereas only 2 of 18 idea-related weaknesses (11.1%) are addressed, suggesting that iterative revision is more effective at improving research execution than altering the core scientific idea.

ing research idea. Our results suggest that autonomous revision can substantially strengthen the execution of existing research, but concerns about novelty and significance generally require returning to the ideation stage. A practical limitation of this work is evaluation cost: running the complete revision process for a single paper, including multiple revision rounds and experiment execution, costs approximately \$65, limiting the number of papers we could evaluate.

References

- Baek, J.; Jauhar, S. K.; Cucerzan, S.; and Hwang, S. J. 2025. ResearchAgent: Iterative Research Idea Generation over Scientific Literature with Large Language Models. *arXiv:2404.07738*.
- Chen, N.; HuiKai, A. L.; Wu, J.; Hou, J.; Zhang, Z.; Wang, Q.; Wang, X.; and He, B. 2026. XtraGPT: Context-Aware and Controllable Academic Paper Revision via Human-AI Collaboration. *arXiv:2505.11336*.
- D’Arcy, M.; Hope, T.; Birnbaum, L.; and Downey, D. 2024. MARG: Multi-Agent Review Generation for Scientific Papers. *arXiv:2401.04259*.
- D’Arcy, M.; Ross, A.; Bransom, E.; Kuehl, B.; Bragg, J.; Hope, T.; and Downey, D. 2024. ARIES: A Corpus of Scientific Paper Edits Made in Response to Peer Reviews. In Ku, L.-W.; Martins, A.; and Srikumar, V., eds., *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 6985–7001. Bangkok, Thailand: Association for Computational Linguistics.
- Ghareeb, A. E.; Chang, B.; Mitchener, L.; Yiu, A.; Szostkiewicz, C. J.; Shved, D.; Gyimesi, G. J.; Laurent, J. M.; Wright, S. M.; Razzak, M. T.; White, A. D.; Finnemann, S. C.; Hinks, M. M.; and Rodriques, S. G. 2026. A Multi-Agent System for Automating Scientific Discovery. *Nature*, 655(8122): 497–505.
- Gottweis, J.; Weng, W.-H.; Daryin, A.; et al. 2026. Accelerating Scientific Discovery with Co-Scientist. *Nature*, 655: 487–496.
- Goyal, P.; Parmar, M.; Song, Y.; Palangi, H.; Pfister, T.; and Yoon, J. 2026. ScholarPeer: A Context-Aware Multi-Agent Framework for Automated Peer Review. *arXiv preprint arXiv: 2601.22638*.
- Idahl, M.; and Ahmadi, Z. 2024. OpenReviewer: A Specialized Large Language Model for Generating Critical Scientific Paper Reviews. *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (System Demonstrations)*.
- Jiang, Y.; and Ng, A. 2026. Stanford Agentic Reviewer. <https://paperreview.ai>. Accessed: 2026-05-24.
- Jin, Y.; Zhao, Q.; Wang, Y.; Chen, H.; Zhu, K.; Xiao, Y.; and Wang, J. 2024. AgentReview: Exploring Peer Review Dynamics with LLM Agents. In *EMNLP*.
- Lu, C.; Lu, C.; Lange, R. T.; Foerster, J.; Clune, J.; and Ha, D. 2024. The AI Scientist: Towards Fully Automated Open-Ended Scientific Discovery. *arXiv preprint arXiv:2408.06292*.
- Luo, Z.; Huang, H.; Lian, J.; Chang, Y.; Xie, X.; and Jin, H. 2026. Can AI Revise Research Papers with Human Review Feedback? An Empirical Study and Benchmark. In *Findings of the Association for Computational Linguistics: ACL 2026*, 17876–17893. Association for Computational Linguistics.
- Mitchener, L.; Yiu, A.; Chang, B.; Bourdenx, M.; Nadolski, T.; Sulovari, A.; Landsness, E. C.; Barabasi, D. L.; Narayanan, S.; Evans, N.; Reddy, S.; Foiani, M.; Kamal, A.; Shriver, L. P.; Cao, F.; Wassie, A. T.; Laurent, J. M.; Melville-Green, E.; Caldas, M.; Bou, A.; Roberts, K. F.; Zagorac, S.; Orr, T. C.; Orr, M. E.; Zvezdaryk, K. J.; Ghareeb, A. E.; McCoy, L.; Gomes, B.; Ashley, E. A.; Duff, K. E.; Buonassisi, T.; Rainforth, T.; Bateman, R. J.; Skarlinski, M.; Rodriques, S. G.; Hinks, M. M.; and White, A. D. 2025. Kosmos: An AI Scientist for Autonomous Discovery. *arXiv:2511.02824*.
- Schmidgall, S.; Su, Y.; Wang, Z.; Sun, X.; Wu, J.; Yu, X.; Liu, J.; Moor, M.; Liu, Z.; and Barsoum, E. 2025. Agent Laboratory: Using LLM Agents as Research Assistants. In Christodoulopoulos, C.; Chakraborty, T.; Rose, C.; and Peng, V., eds., *Findings of the Association for Computational Linguistics: EMNLP 2025*, 5977–6043. Suzhou, China: Association for Computational Linguistics. ISBN 979-8-89176-335-7.
- Tachoyotin, P.; and Acuna, D. 2025. REMOR: Automated Peer Review Generation with LLM Reasoning and Multi-Objective Reinforcement Learning. *arXiv preprint arXiv: 2505.11718*.
- Tan, C.; Lyu, D.; Li, S.; Gao, Z.; Wei, J.; Ma, S.; Liu, Z.; and Li, S. Z. 2024. Peer Review as A Multi-Turn and Long-Context Dialogue with Role-Based Interactions. *arXiv preprint arXiv: 2406.05688*.
- Weng, Y.; Zhu, M.; Bao, G.; Zhang, H.; Wang, J.; Zhang, Y.; and Yang, L. 2025. CycleResearcher: Improving Automated Research via Automated Review. In *The Thirteenth International Conference on Learning Representations*.
- Weng, Y.; Zhu, M.; Xie, Q.; Ning, Z.; Li, S.; Lu, P.; Lin, Z.; Gu, E.; Sun, Q.; and Zhang, Y. 2026. DeepReviewer 2.0:

A Traceable Agentic System for Auditable Scientific Peer Review. *arXiv preprint arXiv: 2604.09590*.

Yamada, Y.; Lange, R. T.; Lu, C.; Hu, S.; Lu, C.; Foerster, J.; Clune, J.; and Ha, D. 2025. The AI Scientist-v2: Workshop-Level Automated Scientific Discovery via Agentic Tree Search. *arXiv preprint arXiv:2504.08066*.

Zhao, B.; Zhang, J.; Whitehouse, C.; Jiang, M.; Shvartsman, M.; Charnalia, A.; Magka, D.; Shavrina, T.; Dunfield, D.; Aodha, O. M.; and Bachrach, Y. 2026. APRES: An Agentic Paper Revision and Evaluation System. *arXiv:2603.03142*.

Zhu, M.; Weng, Y.; Yang, L.; and Zhang, Y. 2025. DeepReview: Improving LLM-based Paper Review with Human-like Deep Thinking Process. *arXiv preprint arXiv: 2503.08569*.